$$= \lim_{h \to 0} -1 = -1$$

$$R[f(x)]_{x=3} = \lim_{x \to 3^+} f(x)$$

$$= \lim_{h \to 0} f(3+h)$$

$$= \lim_{h \to 0} \frac{|(3+h)-3|}{(3+h)-3}$$

$$= \lim_{h \to 0} \frac{|h|}{h}$$

$$= \lim_{h \to 0} 1 = 1$$

NOTE

Here,

$L[f(3)] \neq R[f(3)]$

$\therefore \lim_{x \to 3} f(x)$ does not exist.

Example 5.15

Verify the existence of the function

$$f(x) = \begin{cases} 5x - 4 & \text{if } 0 < x \leq 1 \\ 4x^3 - 3x & \text{if } 1 < x < 2 \end{cases} \quad \text{at } x = 1.$$

Solution

$$L[f(x)]_{x=1} = \lim_{x \to 1^-} f(x)$$

$$= \lim_{h \to 0} f(1-h), \; x = 1 - h$$

$$= \lim_{h \to 0} [5(1-h)-4]$$

$$= \lim_{h \to 0} (1-5h) = 1$$

$$R[f(x)]_{x=1} = \lim_{x \to 1^+} f(x)$$

$$= \lim_{h \to 0} f(1+h), \; x = 1 + h$$

$$= \lim_{h \to 0} [4(1+h)^3 - 3(1+h)]$$

$$= 4(1)^3 - 3(1) = 1$$

Clearly, $L[f(1)] = R[f(1)]$

$\therefore \lim_{x \to 1} f(x)$ exists and equal to 1

NOTE

Let a be a point and f(x) be a function, then the following stages may happen

(i) $\lim_{x \to a} f(x)$ exists but $f(a)$ does not exist.

(ii) The value of $f(a)$ exists but $\lim_{x \to a} f(x)$ does not exist.

(iii) Both $\lim_{x \to a} f(x)$ and $f(a)$ exist but are unequal.

(iv) Both $\lim_{x \to a} f(x)$ and $f(a)$ exist and are equal.

5.2.4 Some results of limits

If $\lim_{x \to a} f(x)$ and $\lim_{x \to a} g(x)$ exists, then

(i) $\lim_{x \to a} (f \pm g)(x) = \lim_{x \to a} f(x) \pm \lim_{x \to a} g(x)$

(ii) $\lim_{x \to a} (fg)(x) = \lim_{x \to a} f(x) \lim_{x \to a} g(x)$

(iii) $\lim_{x \to a} \left(\frac{f}{g}\right)(x) = \frac{\lim_{x \to a} f(x)}{\lim_{x \to a} g(x)}$, whenever $\lim_{x \to a} g(x) \neq 0$

(iv) $\lim_{x \to a} k f(x) = k \lim_{x \to a} f(x)$, where k is a constant.

5.2.5 Indeterminate forms and evaluation of limits

Let $f(x)$ and $g(x)$ are two functions in which the limits exist.

If $\lim_{x \to a} f(x) = \lim_{x \to a} g(x) = 0$, then $\frac{f(a)}{g(a)}$ takes $\frac{0}{0}$ form which is meaningless but does not imply that $\lim_{x \to a} \frac{f(x)}{g(x)}$ is meaningless.

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